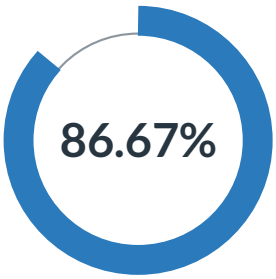
Results

Carter Feng



52
Out of 60 points

28:30
Time for this attempt

Your Answers:

1 0 / 0 points Fill in the Blank

At Seven Lakes High School, we believe that character matters and that academic integrity reflects who we are as learners and citizens. By signing this pledge, I commit to upholding the values of honesty, responsibility, and respect in all academic and personal endeavors.

My Academic Integrity Pledge:

- **I will complete all academic work honestly, without giving or receiving unauthorized assistance.**
- **I will not copy, plagiarize, or submit someone else's work as my own.**
- **I will not use unauthorized materials (notes, devices, online resources) during assessments.**
- **I will respect the integrity of others' work and will not aid others in academic dishonesty.**
- **I will not share or receive academic work, ideas, or answers electronically or in-person when individual work is expected or collaboration is not permitted.**
- **I will report any knowledge of cheating or violations of academic integrity to a trusted adult.**

I understand that academic dishonesty harms my character, undermines trust in our community, and may result in consequences outlined in the school's academic integrity policy.

Print your name to acknowledge the oath. Not printing your name will result in a zero for the assessment.

✓ Carter Feng

2 2 / 2 points Multiple Choice

The **chromatic number** of a graph refers to:

✓ ◉ The minimum number of colors required for proper coloring

○ The number of disconnected components

○ The number of spanning trees

○ The number of cycles in a graph

○ The number of edges in a graph

3 2 / 2 points Fill in the Blank

A graph where every node is connected to every other node is called a ✓ complete graph.

4 2 / 2 points Multiple Choice

The time complexity of BFS and DFS is:

○ O(log V)

○ $O(V^2)$

○ O(V)

○ O(E)

✓ ◉ O(V+E)

5 2 / 2 points Multiple Choice

The space complexity of an adjacency **matrix** for a graph with V nodes is:

- ○ O(1)
- ✓ ◉ $O(V^2)$
- ○ O(V+E)
- ○ O(E)
- ○ O(V)

6 2 / 2 points Multiple Choice

1. A **tree** is a special type of graph that:

- ✓ ◉ Has exactly n−1 edges for n nodes
- ○ Has all nodes with the same degree
- ○ Is always weighted
- ○ Has more edges than nodes
- ○ Contains at least one cycle

7 2 / 2 points Multiple Choice

Which of the following is true about Depth-First Search (DFS)?

- ○ It always finds a solution
- ○ It guarantees the shortest path
- ○ It uses a queue

- ✓ It explores all children before siblings
- It is optimal for shortest path searches

8 2 / 2 points Multiple Choice

A **bipartite graph** can always be

- A weighted graph
- Drawn as a planar graph
- ✓ Colored using two colors
- Cyclic

9 2 / 2 points Multiple Choice

A graph with an odd-length cycle cannot be:

- Planar
- Weighted
- ✓ Bipartite
- Directed
- Connected

10 2 / 2 points Multiple Choice

The main **advantage** of an adjacency matrix is:

- It always has fewer edges
- It is efficient for sparse graphs

- It is better for hierarchical data
- It uses less space
- ✓ It is efficient for dense graphs

11 2 / 2 points Multiple Choice

In an **undirected graph**, an edge between two nodes allows movement:

- In one direction only
- Only if weighted
- Only if there is a cycle
- ✓ In multiple directions
- Only if adjacent

12 2 / 2 points Multiple Choice

A graph that has no edges is called a:

- ✓ Empty Graph
- Weighted Graph
- Bipartite Graph
- Complete Graph
- Directed Graph

13 2 / 2 points Multiple Choice

The **best representation** for a sparse graph is:

✓ Adjacency List

Hash Table

Incidence Table

Edge Table

Adjacency Matrix

14 2 / 2 points Multiple Choice

The degree of a node refers to:

The weight of its edges

The depth of the node

✓ The number of edges connected to it

The total number of nodes in the graph

The number of nodes it connects to

15 2 / 2 points Multiple Choice

A **complete graph** is defined as:

A weighted graph

A graph with cycles

A graph with no edges

✓ A graph where every node is connected to every other node

A disconnected graph

16 2 / 2 points Multiple Choice

What are the two main components of a graph?

- Edges and Cycles
- Components and Trees
- ✓ Nodes and Edges
- Nodes and Paths
- Vertices and Degrees

17 2 / 2 points Categorization

What do you do with the following power plant cards?

Starts in the market

- ✓ Power Plant 5

Starts in the futures market

- ✓ Power Plant 7

Placed on the top of the deck

- ✓ Power Plant 13

Placed on the bottom of the deck

- ✓ Power Plant Step 3

Shuffled in deck

- ✓ Power Plant 11

Does not exist

- ✓ Power Plant 2
- ✓ Power Plant Set 2

18 2 / 2 points Multiple Choice

At the beginning of the game, how much coal is place in the market?

- 2 coal is placed in the market at the $14 box and moving to the right.
- 6 coal is placed in the market at the $7 box and moving to the right
- ✓ all the coal
- all but 6 coal is placed in the market, starting in the $3 box and moving to the right

19 2 / 2 points Multiple Choice

Who wins the game

- ✓ The player that can power the most cities, then most money, then most cities connected
- The player with the most cities connected, then the most powered cities, then the most money.
- Stroud, always Stroud
- The player with the most money, then the most powered cities, then the most cities connected

20 2 / 2 points Fill in the Blank

There are a total of ✓ 24 tokens each representing coal, oil, and garbage resources.

21 0 / 2 points Multiple Choice

When can a person buy into a city owned by 2 players?

- After Step 3 begins

✗ ◉ After Step 2 begins

Correct Answer: **After Step 3 begins**

○ They can always do it

22 0 / 2 points Multiple Choice

The upper left hand corner of the power plant has a number between 3 and 50.

What does it represent

○ Which player will go first if they are tied at the same number of cities

○ The number of cities required to own such that the power plant is removed from the auction house

○ All of these answers are correct

✗ ◉ The minimum amount that can be bid at auction for the power plant

Correct Answer: **All of these answers are correct**

23 2 / 2 points True or False

A player must buy a power plant on every round.

○ True

✓ ◉ False

24 0 / 2 points Multiple Choice

Once the Auction Power Plants Phase is complete, reorganize the playing order.

○ false

✗ true

Correct Answer: **only in the first round**

only in the first round

25 2 / 2 points Multiple Choice

What is the special rule for a 4 player game?

The game will end when a player connects 15 cities.

Remove 8 power plants from the game (but none of the starting plants)

The game will end when a player connects 21 cities.

✓ Remove 4 power plants from the game(but none of the starting plants)

26 2 / 2 points Multiple Choice

The power plant shows a house with a number in it. What does that number represent?

The number of cities the player may buy on the board.

The minimum number of cities a player needs to own to use the power plant

✓ The number of cities the power plant can power on a given turn

The number of cities the power plant can power with each set of resources

27 0 / 2 points Multiple Choice

When does Step 3 begin?

During the Buying Material phase, when the Step 3 power plant card is drawn during the Auction Phase.

✗ At the start of the next turn, when the Step 3 is drawn in the Bureaucracy phase

Correct Answer: **Each one is correct.**

- Each one is correct.
- During Bureaucracy phase, when the Step 3 is drawn during the Building Phase

28 2 / 2 points Multiple Choice

How does the player order go when in the Buying Resources Phase of the game?

- From 1 to N, where N is the number of players
- From Player N, where N is the number of players, and then clockwise around the table.
- From Player 1 in the playing order and then clockwise around the table
- ✓ From N to 1, where N is the number of players

29 2 / 2 points Multiple Choice

How does the game end?

- When someone can provide energy to at least 17 cities in a 3 or 4 player game, 15 cities in a 5 player game
- ✓ When someone connects 17 cities in a 3 or 4 player game, 15 cities in a 5 player game
- When all receive electricity
- When all power plant cards are gone.

30 2 / 2 points True or False

During The buying resource phase, a player may purchase any resource on market for the listed price as long as that player has the storage capacity to hold the resource.

- [x] True
- [ ] False

31 2 / 2 points Multiple Choice

How much does the first city a player buys in the game cost the player?

- [ ] 20
- [ ] 20 + connecting cost
- [x] 10
- [ ] 10 + connecting cost
- [ ] 15
- [ ] 15 + connecting cost